GMS1000 Project Proposal

Alleviating Chronic Low-Grade Inflammation for Increased Immune Response to Vaccines in the Elderly

Submitted by:

Group 3

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ABSTRACT

Vaccines are one of the most significant medical advancements in human history, playing a vital role in protecting individuals and communities from infectious diseases. Through the use of vaccines, many infectious diseases have been eradicated or reduced significantly, resulting in improved health outcomes and decreased mortality rates worldwide. The elderly population is at a higher risk of developing severe complications from infectious diseases, leading to the prioritisation of vaccinations in this age group. However, several factors, such as age-related changes in the immune system, coexisting medical conditions, and medications, can decrease vaccine efficacy in the elderly. Through measuring and comparing cytokine and TNF-α levels, this study discusses the processes by which immunosenescence and inflammaging reduces immune response and vaccine efficacy in the elderly. Subsequently, the manipulation of TNF-α is explored as a potential strategy to improve vaccine efficacy in the elderly.

1. BACKGROUND AND CLINICAL SIGNIFICANCE

1.1 Ageing and Immunosenescence

Human life expectancy has doubled since 1800 (Finch, 2010). This drastic increase is largely attributed to advancements in medical care and a higher standard of living. With an extended lifespan and a rising proportion of the elderly, the association of ageing with an increased predisposition to chronic morbidity and frailty has emerged as a major public health concern (Ferrucci & Fabbri, 2018).

The gradual deterioration in physiological functions is a common sign of ageing. Immunosenescence, the dysregulation of the immune system, is one of the main characteristics of physiological function deterioration (Pereira et al., 2020). Immunosenescence negatively alters both innate and adaptive immune responses to infections by diminishing the number and function of immune cells (Nguyen et al., 2015), as well as the involution of primary lymphoid organs. This shrinkage depletes the pool of naive T and B cells available for an adaptive immune response. Alterations to the functionality of immune cells also affect antigen presentation due to reduced antigen uptake, decreased phagocytic function and aberrant cytokine production (Pereira et al., 2020). Therefore, immunosenescence can have implications in vaccine-induced protection against infectious diseases. As vaccine efficacy is largely dependent on the quality of the immune system, immunocompromised individuals such as the elderly are often not adequately protected despite immunisation (Weinberger et al., 2008). A study conducted by Goodwin et. al. revealed that against influenza virus strains, persons above 74 years old only had a seroprotection of 29%-46% while persons aged 60-74 demonstrated a higher seroprotection at 41%-58% (2006). Consequently, this data implies that antibody response deteriorates with age and can have serious implications for vaccination, especially against emerging infectious diseases.

1.2 Inflammaging

While it is largely known that immunosenescence forms the basis of reduced vaccine efficacy, another pervasive feature accompanying the ageing process that may contribute to a less efficacious vaccine response is inflammaging (Pereira et al., 2020). Inflammaging is defined as a chronic and systemic sterile pro-inflammatory status

associated with the elderly (Pereira et al., 2020). It occurs in the absence of overt infections and presents as a low-grade inflammation (Franceschi & Campisi, 2014). Epidemiological studies have shown that inflammaging is predictive of morbidity and earlier mortality in the elderly as it contributes to a vast range of age-related diseases such as cardiovascular disease, chronic kidney disease and diabetes mellitus (Ferrucci & Fabbri, 2018). This state of chronic low-grade inflammation is supported by clinical evidence showing higher circulating levels of pro-inflammatory markers such as tumour necrosis factor (TNF), interleukin-6 (IL-6) and other pro-inflammatory cytokines in the serum of the elderly as opposed to younger individuals (Pereira et al., 2020).

Beyond the defects of the immune system caused by immunosenescence, inflammaging may also have a role in causing a decreased immune response to vaccines (Pereira et al., 2020). While there have been studies suggesting that excessive baseline inflammation is associated with poor vaccine responses (Pereira et al., 2020), the aetiology of inflammaging and its potential impacts on vaccination still remain largely unknown. Further work on the full spectrum of changes pertaining to inflammaging and ageing is fundamental to implementing better strategies for improving vaccine efficacy (Pereira et al., 2020).

1.2.1 Cellular Senescence as a Main Cause of Inflammaging

In addition to the reduced ability of the aged immune system to eliminate pathogens (Weinberger et al., 2008), a multitude of factors such as cell debris, pro-coagulation factors and gut dysbiosis causing microbial translocation have been shown to contribute to inflammaging in the elderly (Sanada et al., 2018). However, one of the major persistent causes of prolonged inflammaging is cell senescence, which refers to the irreversible cell cycle arrest resulting in the cessation of cellular growth (Sanada et al., 2018). Increased age is correlated to an increased number of senescent cells acquiring the senescence-associated secretory phenotype (SASP) in several organs (Sanada et al., 2018). These cells with the SASP are known to secrete high levels of pro-inflammatory cytokines and chemokines, contributing to the low-grade inflammation and further immune dysregulation observed in the elderly.

2 HYPOTHESIS AND SPECIFIC AIMS

Inflammation is often indicative of a responsive immune system. However, since the elderly are always in a constant state of low-grade inflammation caused by inflammaging, it has been speculated that this chronic inflammatory process may impede the ability of the aged immune system to trigger an optimal immune response to the antigens and stimuli induced by vaccines. Inflammaging could have increased the critical threshold for immune responses and hence, elderly with inflammaging may not respond to a low load of antigens in vaccines. It is hypothesised that controlling or reducing this chronic low-grade inflammation will yield an increased vaccine response and production of higher titers of antibodies in the elderly.

Thus, the first aim is to establish a link between inflammaging and vaccine response by analysing PBMCs derived from the elderly (above 65 years old) and adults (21 to 65 years old). The vaccines that will be used are Flucelvax Quadrivalent and Flublok Quadrivalent vaccines. Flucelvax Quadrivalent is an inactivated vaccine (Center for Biologics Evaluation and Research, 2020) while Flublok Quadrivalent is a recombinant DNA-based vaccine (Sanofi, n.d.). Both vaccines are approved by the United States Food Drug Administration for administration to people of all ages during the 2022-2023 flu season. We can later deduce whether inflammaging is present amongst the elderly and the cause of a decreased vaccine response.

Tumour necrosis factor (TNF) is a protein that induces inflammation (Laha et al., 2021). While inflammation is an important aspect and indicator of an immune response, excessive amounts of TNF-α can lead to a slew of health complications such as chronic inflammation, which may eventually lead to the development of pathological complications such as autoimmune diseases (Jang et al., 2021). As such, anti-TNF therapy is commonly used to reduce inflammation as it blocks TNF action (Rossi et al., 2015). Therefore, we propose the use of infliximab, an anti-TNF drug, to reduce inflammation and thus increase vaccine efficacy.

To understand the role of infliximab in potentially increasing vaccine efficacy, we sought to characterise vaccine response in the elderly following infliximab and placebo treatment prior to receiving their vaccination. Subsequently, we will be able to deduce

whether anti-TNF drugs such as infliximab can be used to increase vaccine efficacy amongst the elderly.

3 METHODS AND APPROACH

To study the first aim of the study, blood samples from 50 healthy adult Singaporeans between ages 21- 65 and 50 elderly Singaporeans above 65 years old will be collected to obtain peripheral blood mononuclear cells (PBMCs) using SepMate PBMC isolation tubes (Figure 1). The PBMCs from both study groups will either be left untreated or treated with Flucelvax Quadrivalent and Flublok Quadrivalent influenza vaccine for 24 hours. The PBMCs treated with influenza vaccines from both study groups should secrete proinflammatory cytokines like interferon-gamma (IFNγ), IL-10 or IL-6 (Bernstein et al., 1998). The supernatant of these PBMCs will be harvested after 24 hours for enzyme-linked immunosorbent assay (ELISA) and be screened for IFNγ, IL-10 and IL-6. It is expected that these three cytokines will have a lower increase in levels in influenza vaccine-treated PBMCs compared to the untreated PMBCs in the Singaporean elderly as compared to the Singaporean adults. Hence, the expected results of the ELISA should show that there is a weaker immune response to both influenza vaccines in the Singaporean elderly than in the Singaporean adults, thus indicating the presence of inflammaging.

To investigate whether infliximab usage can be used to increase vaccine efficacy amongst the elderly, PBMCs from the same 50 elderly Singaporeans will be used (Figure 1). These 50 elderly will be equally divided into 2 groups of 25 individuals by random assignment. The PBMC samples from Group 1 will be pre-treated with 0.5mg of TNF inhibitor infliximab whereas the PBMC samples from Group 2 will be pre-treated with 0.5mg of IgG2a, an isotype control. These samples will be pre-treated for a duration of 1 hour. After pre-treatment, cell wash will be performed to remove residual drugs before screening the samples for the same panel of 3 cytokines via ELISA to establish their baseline levels prior to introducing the vaccines. Once established, Flucelvax Quadrivalent and Flublok Quadrivalent influenza vaccines will be introduced to the samples in both groups. Cytokine production has been documented to peak between 24 and 48 hours in response to stimuli (Sullivan et al., 2000). IFN-γ, IL-6 and

IL-10 cytokine levels will be assessed periodically at intervals of 8 hours to monitor the immune response to the vaccines added. It is hypothesised that following vaccination, the samples pre-treated with infliximab should exhibit a greater increase in the concentrations of all 3 cytokines as compared to the samples treated with the placebo. Each cytokine level will be charted graphically and monitored over a time period of 48 hours for each sample in each group. The mean concentration values of each cytokine will be computed for each group. A statistical test will be performed to determine if any differences in the mean values between the 2 groups are significant. This will enable us to conclude if infliximab usage is associated with an increased vaccine response and hence, provide evidence to suggest that anti-TNF drugs can be employed as a potential strategy in eliciting a more robust immune response to vaccines in the elderly.

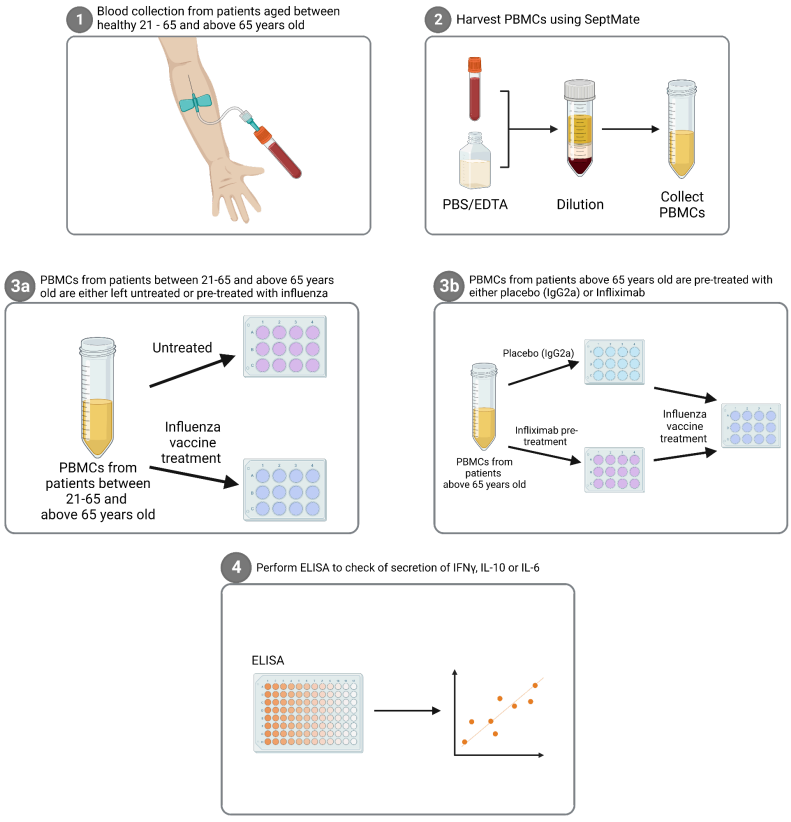


Figure 1: Schematic diagram of the experimental procedures to be done to establish a link between inflammaging and vaccine response. Steps 3a and 3b shows the method to answer Aim 1 and 2 respectively. Created with BioRender.com.

3.1 Limitations

However, a few limitations of the experiment have been identified. Firstly, it is worthy to note that the use of PBMCs *in vitro* may not accurately reflect the *in vivo* environment of the body after vaccination because inside the body, the immune cells are exposed to a range of different stimuli and immunological niche that could potentially influence the response to the vaccine. Nonetheless, the data will provide insights into how immune cells respond to vaccines. If the outcome is positive, future work can be performed to test the effect of infliximab in animal models, and eventually in clinical trials. Secondly, as PBMCs are a mixed population of different cells, we were unable to identify the specific cells involved in the host response. Future work using single cell sequencing or Cytometry by Time-Of-Flight (CYTOF) can help elucidate the contributions.

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